MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 08** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. The distance between two parallel tangents to a circle of radius 4.5 cm is: [1]

(A) 4.5 cm (B) 9 cm (C) 13.5 cm (D) 18 cm

Q2. In quadrilateral $ABCD$ circumscribing a circle, $AB = 7$ cm, $BC = 8.5$ cm, and $CD = 5$ cm. The length of DA is: [1]

(A) 3.5 cm (B) 4.5 cm (C) 6.5 cm (D) 5.5 cm

Q3. If two concentric circles have radii 25 cm and 7 cm, the length of the chord of the larger circle touching the smaller circle is: [1]

(A) 24 cm (B) 36 cm (C) 48 cm (D) 50 cm

Q4. If PT is a tangent at T to a circle with centre O and $\triangle OPT$ is isosceles with $OP = \sqrt{2} \cdot OT$, then $\angle POT$ is: [1]

(A) $30°$ (B) $45°$ (C) $60°$ (D) $90°$

Q5. The common point of a tangent to a circle and the circle is called the: [1]

(A) Centre (B) Point of contact (C) Origin (D) Vertex

Q6. **Assertion (A):** The opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.
Reason (R): Tangents drawn from an external point to a circle are equal in length. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. If $\angle PTQ = 80°$, find the measure of $\angle OPQ$. [2]

Q8. Find the inradius of a right-angled triangle whose base is 15 cm and hypotenuse is 17 cm. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. If PA and PB are tangents to a circle with centre O such that $\angle APB = 60°$ and radius $OA = 3$ cm, find the length of OP and the length of each tangent. [3]

Q10. A circle is inscribed in a right $\triangle ABC$ with $\angle B = 90°$, $BC = 12$ cm and $AB = 5$ cm. Find the area of the portion of the triangle that lies outside the circle. (Use $\pi = 3.14$) [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

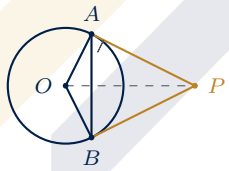
Q11. Prove that the parallelogram circumscribing a circle is a rhombus. Hence, if one of the interior angles of this circumscribing parallelogram is 90°, prove that it must be a square. [5]

Q12. PQ is a chord of length 16 cm of a circle of radius 10 cm. The tangents at P and Q intersect at a point T. Find the length of the tangent TP and the area of $\triangle TPQ$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Laser Alignment across Circular Optical Cavities

In an optical laboratory, a laser beam emitter at point P directs two beams PA and PB tangent to a circular optical resonator of radius $r = 3$ cm centred at O. The emitter is placed at distance $OP = 6$ cm.



(i) Find the angle $\angle APO$ between the emitter line and tangent beam PA. [1]

(ii) Calculate the total angle of beam divergence $\angle APB$. [1]

(iii) Find the length of each laser beam PA, OR determine the length of the chord of contact AB. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★